A look into the weather in Tromsø over time

There has been a lot of talk this winter season about the weather and how it has changed over the years. This made me curious about how much the weather actually has changed and if the observations reflect what people are feeling. In this project we will then look at a couple of aspects described in the section below.

Hypothesis

This project is split into different sections with different focuses. To make it easier to follow, there are 3 proposed hypothesis that this project will try to answer. They are the following;

1. The temperature in Tromsø has increased, with warmer summers and winters
2. The weather has become more extreme, with higher temperatures and stronger winds classified as extreme (top 5%)
3. There exists some local differences between the different sensors, especially depending on the geography

Importing and reading nessecary files

Includes nessesary libraries as well as some global variables and changes that needs to be included for the code in this program to run

In [1]:
```
#Importing of needed libraries

import numpy as np
import pandas as pd
import geopandas as gpd

import matplotlib.pyplot as plt
import altair as alt
import reading_data as rd
```

In [2]:
```
client_id = '../client_files/client_id.txt'
client_secret = '../client_files/client_secret.txt'
alt.data_transformers.disable_max_rows()
```

Out[2]:
```
DataTransformerRegistry.enable('default')
```

Finding the weather stations of Tromsø

Before startning we need to figure out where the weather stations are in Tromsø and for how long have they been active. This can help inform what observational data to look at and what aspect of time we are dealing with.

In [3]:
```
#Finding the station IDs and names for the different weather stations in
station_id = pd.DataFrame(rd.FindingStations(client_id, client_secret
                                            ).get_sensors(69.95, 18.95,
                                            columns=['station_id', 'name

#Using this information to collect general information about the differen
start_date, coor, names = [], [], []
for station in station_id['station_id']:
    location = rd.ReadingData(station, client_id, client_secret)
    start, end, coordinates, name = location.get_station_info()
    start_date.append(start)
    coor.append(coordinates)
    names.append(name)
```

In [4]:
```
#Making the information easier to read and prepare it for plotting later
station_id['date'] = start_date
station_id['year'] = pd.to_datetime(station_id['date']).dt.strftime('%Y')
station_id['lon'] = [cor[0] for cor in coor]
station_id['lat'] = [cor[1] for cor in coor]
station_id['name'] = names
station_id.sort_values(by='date', inplace=True)
station_id.reset_index(drop=True, inplace=True)
station_id
```

Out[4]:

	station_id	name	date	year	lon	lat
0	SN90450	Tromsø (Vervarslinga)	1895-08-01	1895	18.93680	69.65370
1	SN90490	Tromsø LH	1964-09-30	1964	18.91330	69.67670
2	SN90400	Tromsø (Holt)	1987-05-04	1987	18.90950	69.65380
3	SN90491	Stor-Kjølen	1998-04-02	1998	18.78750	69.73330
4	SN90720	Måsvik	2003-10-08	2003	18.69380	69.99030
5	SN91020	Breivikeidet	2003-12-05	2003	19.51030	69.63670
6	SN90980	Oldervik	2003-12-05	2003	19.66470	69.75650
7	SN90721	Gråhaugen	2005-08-04	2005	18.70530	69.99970
8	SN90495	Stakkevollan	2011-10-20	2011	18.98200	69.69420
9	SN90451	Tromsø PLU	2011-10-20	2011	18.93650	69.65370
10	SN90510	Tromsdalen	2011-10-20	2011	19.00650	69.65080
11	SN90560	Kvaløysletta	2011-10-20	2011	18.87720	69.69880
12	SN90360	E8 Sandvikeidet	2013-06-11	2013	19.00867	69.54633
13	SN90612	Damvatnet	2017-10-03	2017	19.02980	69.83680
14	SN91010	Fv91 Breivikeidet	2019-02-21	2019	19.56783	69.65600
15	SN90480	E8 Breivika	2024-11-11	2024	18.96923	69.67677
16	SN90295	Sessøya	2025-01-14	2025	18.25000	69.75000
17	SN90290	Angstaursundet	2025-01-14	2025	18.17090	69.69950

In [5]:
```
#Creating a GeoDataFrame with the coordinates of the weather stations and
station_coordinates = gpd.GeoDataFrame(station_id,
```

```
                                       geometry=gpd.points_from_xy(
                                           station_id.lon, station_id.lat
station_coordinates.crs = 'EPSG:4326'
station_coordinates.head(20)
```

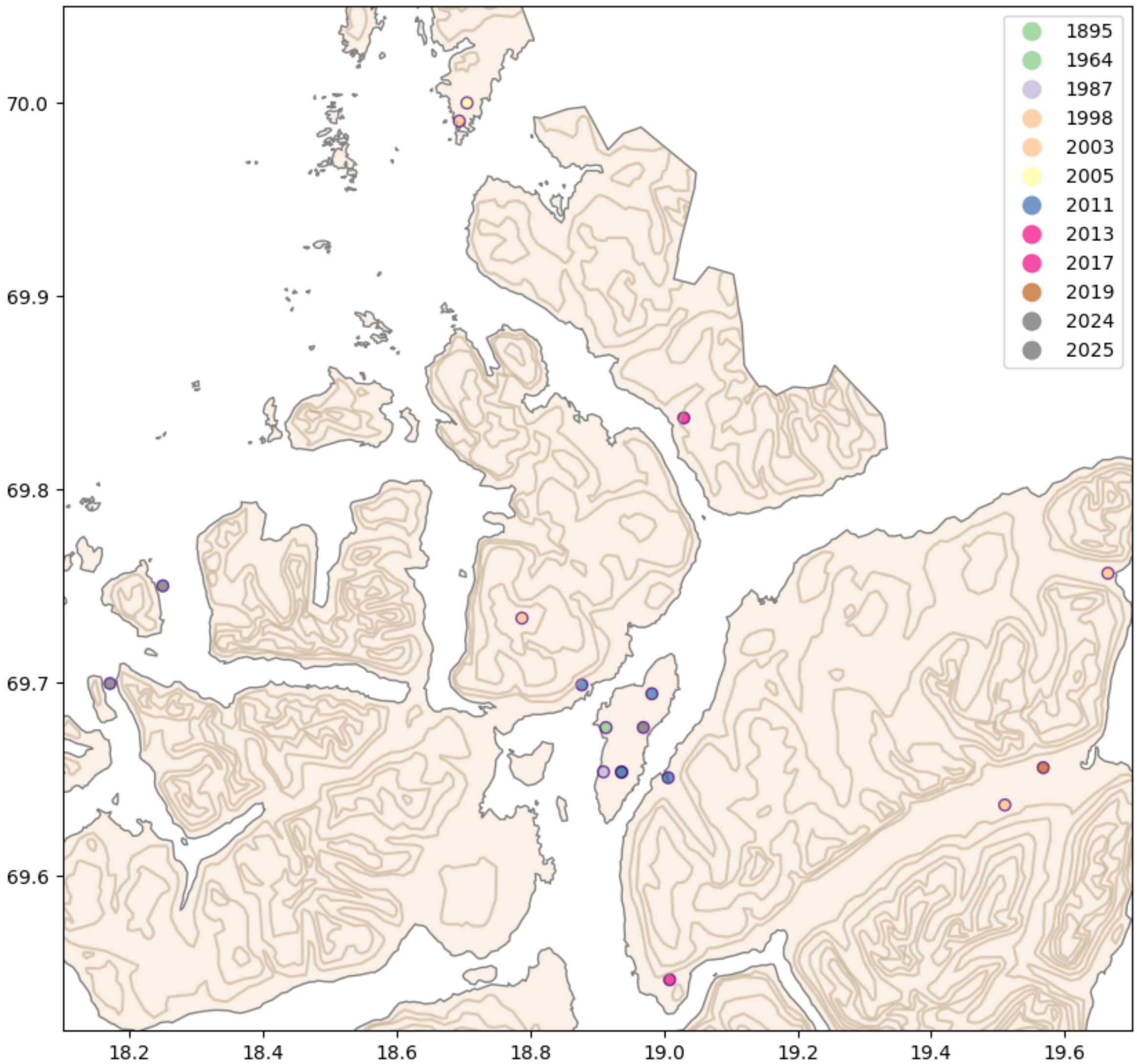
Out [5]:

	station_id	name	date	year	lon	lat	geometry
0	SN90450	Tromsø (Vervarslinga)	1895-08-01	1895	18.93680	69.65370	POINT (18.9368 69.6537)
1	SN90490	Tromsø LH	1964-09-30	1964	18.91330	69.67670	POINT (18.9133 69.6767)
2	SN90400	Tromsø (Holt)	1987-05-04	1987	18.90950	69.65380	POINT (18.9095 69.6538)
3	SN90491	Stor-Kjølen	1998-04-02	1998	18.78750	69.73330	POINT (18.7875 69.7333)
4	SN90720	Måsvik	2003-10-08	2003	18.69380	69.99030	POINT (18.6938 69.9903)
5	SN91020	Breivikeidet	2003-12-05	2003	19.51030	69.63670	POINT (19.5103 69.6367)
6	SN90980	Oldervik	2003-12-05	2003	19.66470	69.75650	POINT (19.6647 69.7565)
7	SN90721	Gråhaugen	2005-08-04	2005	18.70530	69.99970	POINT (18.7053 69.9997)
8	SN90495	Stakkevollan	2011-10-20	2011	18.98200	69.69420	POINT (18.982 69.6942)
9	SN90451	Tromsø PLU	2011-10-20	2011	18.93650	69.65370	POINT (18.9365 69.6537)
10	SN90510	Tromsdalen	2011-10-20	2011	19.00650	69.65080	POINT (19.0065 69.6508)
11	SN90560	Kvaløysletta	2011-10-20	2011	18.87720	69.69880	POINT (18.8772 69.6988)
12	SN90360	E8 Sandvikeidet	2013-06-11	2013	19.00867	69.54633	POINT (19.00867 69.54633)
13	SN90612	Damvatnet	2017-10-03	2017	19.02980	69.83680	POINT (19.0298 69.8368)
14	SN91010	Fv91 Breivikeidet	2019-02-21	2019	19.56783	69.65600	POINT (19.56783 69.656)
15	SN90480	E8 Breivika	2024-11-11	2024	18.96923	69.67677	POINT (18.96923 69.67677)

	station_id	name	date	year	lon	lat	geometry
16	SN90295	Sessøya	2025-01-14	2025	18.25000	69.75000	POINT (18.25 69.75)
17	SN90290	Angstaursundet	2025-01-14	2025	18.17090	69.69950	POINT (18.1709 69.6995)

In [6]:

```
#Plotting how old and where the different weather stations are in Tromsø
#The source for the underlying map can be found in the sources section of
gdf = gpd.read_file('../data/Kommuner-L.geojson')
hoyde = gpd.read_file('../data/Basisdata_55_Troms_25833_N1000Hoyde_GML.gm
                      layer='Høydekurve')
hoyde = hoyde.to_crs(gdf.crs)
tromso = gdf[gdf['name'] == 'Tromsø']
clipped_hoyde = gpd.clip(hoyde, tromso)
fig, ax = plt.subplots(figsize=(10, 10))
tromso.plot(ax=ax, color="linen", edgecolor="grey")
clipped_hoyde.plot(ax=ax, color="xkcd:taupe", markersize=0.1, alpha=0.5)
station_coordinates.plot(ax=ax, column='year', cmap='Accent', marker='o',
          alpha=0.7, edgecolor='indigo')
ax.set_xlim(18.1,19.7)
ax.set_ylim(69.52,70.05)
plt.show()
```



There is a nice variety of places and ages of the different weather stations in Tromsø, with the oldest being 130 years this year. Criterias for good stations have been set as a variety of locations geographically and at least 20 years of data. Looking at our criteria we then end up with the following list

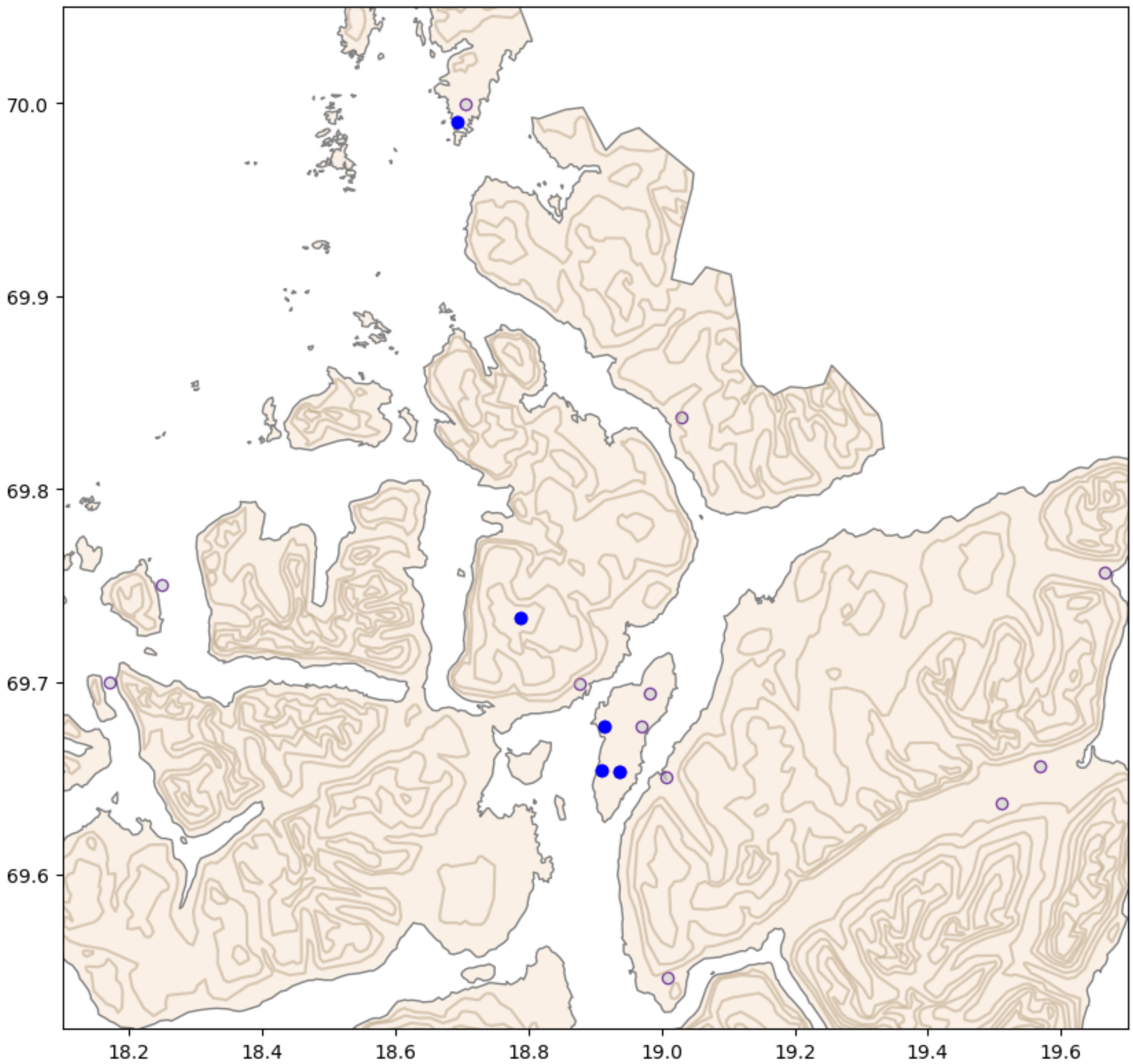
In [7]:
```
chosen_stations = station_coordinates.iloc[[0, 1, 2, 4, 3]]
chosen_stations
```

Out[7]:

	station_id	name	date	year	lon	lat	geometry
0	SN90450	Tromsø (Vervarslinga)	1895-08-01	1895	18.9368	69.6537	POINT (18.9368 69.6537)
1	SN90490	Tromsø LH	1964-09-30	1964	18.9133	69.6767	POINT (18.9133 69.6767)
2	SN90400	Tromsø (Holt)	1987-05-04	1987	18.9095	69.6538	POINT (18.9095 69.6538)
4	SN90720	Måsvik	2003-10-08	2003	18.6938	69.9903	POINT (18.6938 69.9903)
3	SN90491	Stor-Kjølen	1998-04-02	1998	18.7875	69.7333	POINT (18.7875 69.7333)

We will mainly use the first one named Tromso (Vervarslinga) as that is where we have the most data from, however when comparing locally we will try to use the remaining ones in the list. To give an overview of where they are, they are highlighted in the map shown under.

In [8]:
```
#Plotting the same as earlier, but highlighting the stations above
fig, ax = plt.subplots(figsize=(10, 10))
tromso.plot(ax=ax, color="linen", edgecolor="grey")
clipped_hoyde.plot(ax=ax, color="xkcd:taupe", markersize=0.1, alpha=0.5)
station_coordinates.plot(ax=ax, color='lightgrey', marker='o', legend=Tru
          alpha=0.7, edgecolor='indigo')
chosen_stations.plot(ax=ax, color='Blue', marker='o', legend=True)
ax.set_xlim(18.1,19.7)
ax.set_ylim(69.52,70.05)
plt.show()
```



1. The temperature in Tromsø has increased, with warmer summers and winters

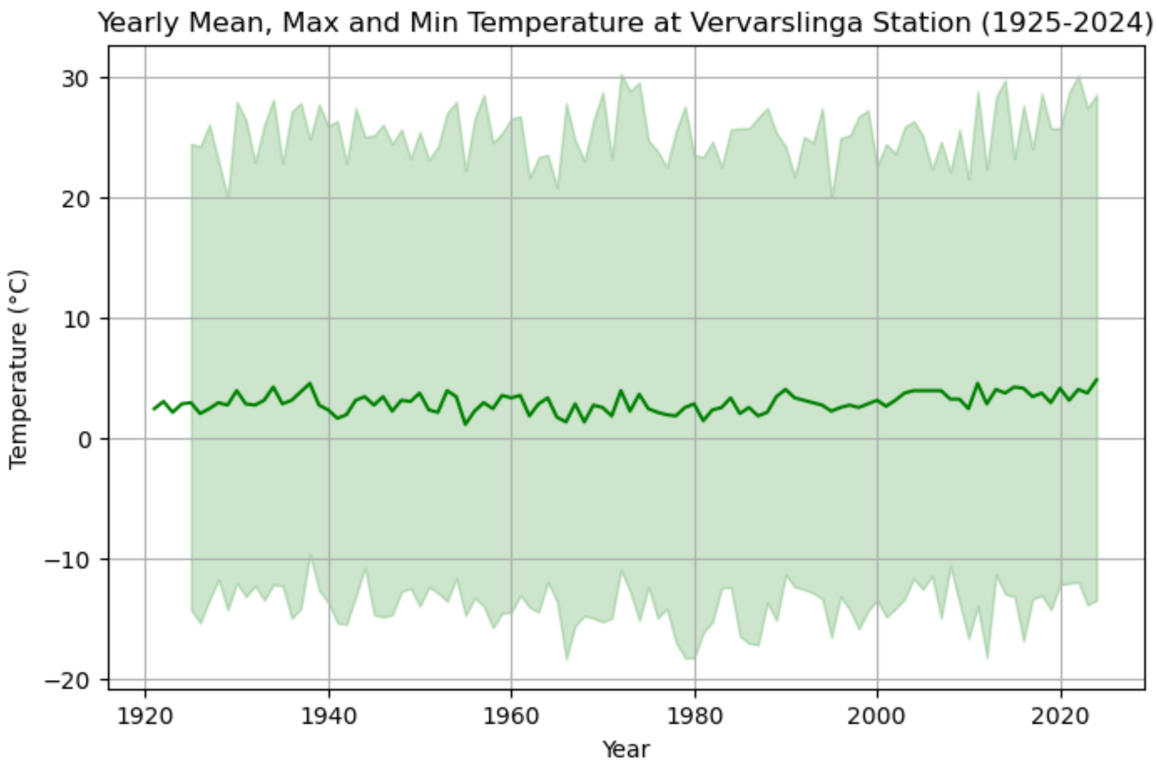
According to NASA 2024 was the warmest year on record, but how much warmer is it in Tromsø? How much has the seasons change?

In [9]:
```
#Collecting the yearly temperature data from the Vervarslinga staion from
vervarslinga = rd.ReadingData('SN90450', client_id, client_secret)
temperature_yearly = vervarslinga.get_temperature_yearly('1885','2024')
```

First instinct is to look at how the yearly temperatures in Tromsø has changed. To account for summer and winter, the max, min and average is collected and then plotted under

In [31]:
```
plt.figure(figsize=(8, 5))
plt.plot(temperature_yearly['year'], temperature_yearly['mean'], linestyl
plt.fill_between(temperature_yearly['year'], temperature_yearly['min'], t

# Labels and title
plt.xlabel("Year")
plt.ylabel("Temperature (°C)")
plt.title("Yearly Mean, Max and Min Temperature at Vervarslinga Station (
```

```
plt.grid(True)
plt.show()
```



The plot does hint to a trend going upwards in temperature, however it isn't a very strong trend. This could be because we are plotting over a bigger range, however plots showing only one of the values doesn't show much either. This could also be because of the yearly values and having more detailed observations like daily will give a better picture at how things have changed in Tromsø over time. A note is also that there isn't any good data before 1925, so while we do have some observational data from before that, it is not usable for this project.

Splitting the data into the seasons when collecting the observations will make it easier to look at winter and summer.

In [11]:
```
#Getting all the temperature data for the spring
spring_temperature = pd.DataFrame()
for year in range(1925, 2025):
    spring_temperature = pd.concat([spring_temperature, vervarslinga.get_
```

In [12]:
```
#Getting all the temperature data for the summer
summer_temperature = pd.DataFrame()
for year in range(1925, 2025):
    summer_temperature = pd.concat([summer_temperature, vervarslinga.get_
```

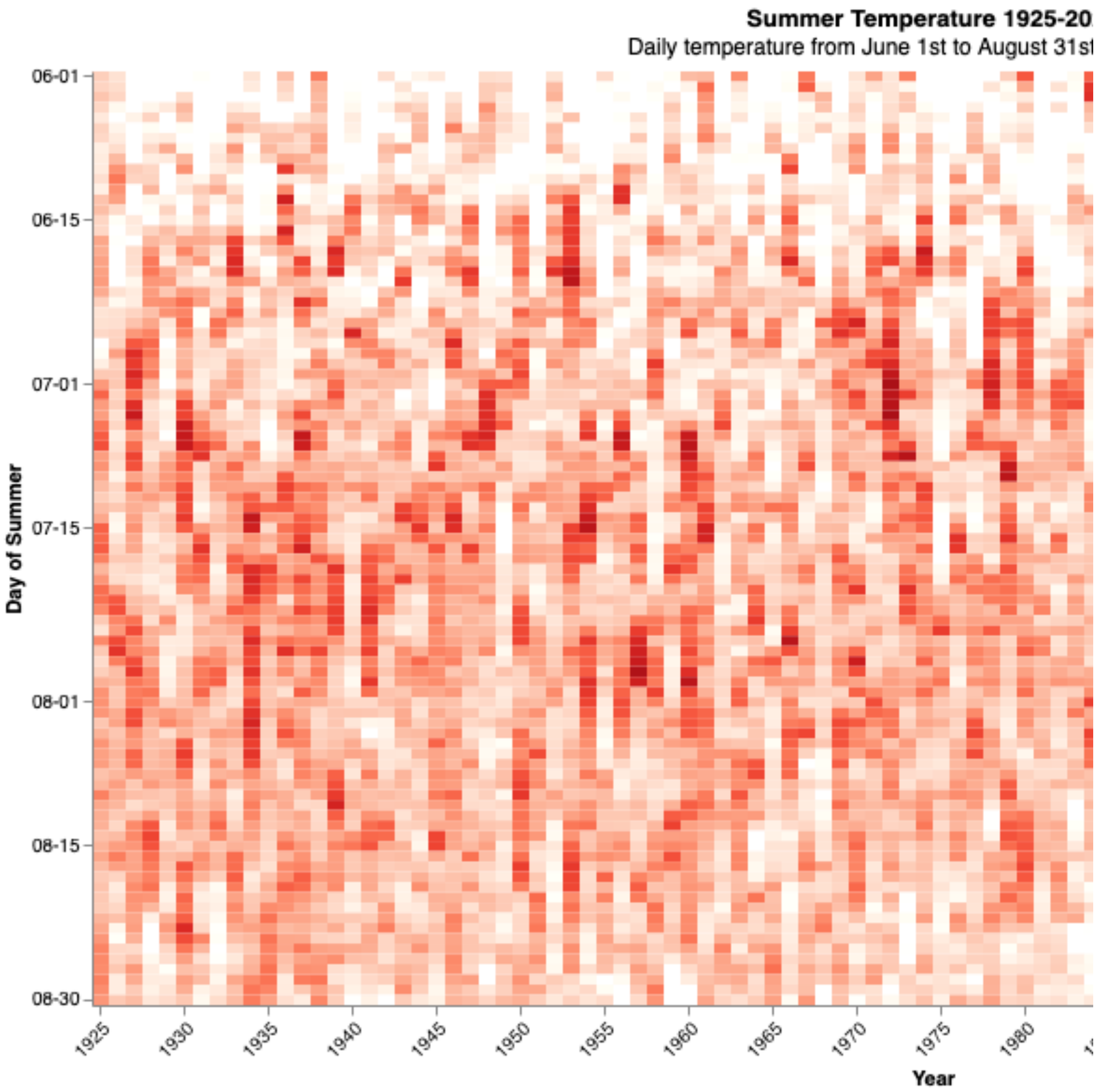
In [13]:
```
#Getting all the temperature data for the autumn
atumn_temperature = pd.DataFrame()
for year in range(1925, 2025):
    atumn_temperature = pd.concat([atumn_temperature, vervarslinga.get_te
```

In [14]:
```
#Getting all the temperature data for the winter
winter_temperature = pd.DataFrame()
for year in range(1925, 2025):
    winter_temperature = pd.concat([winter_temperature, vervarslinga.get_
```

In [15]:
```
#Reformatting the dataframes to make them easier to work with
for frames in [spring_temperature, summer_temperature, atumn_temperature,
    frames['date'] = pd.to_datetime(frames['date'])
    frames['year'] = frames['date'].dt.year
    frames['month_day'] = frames['date'].dt.strftime("%m-%d")
```

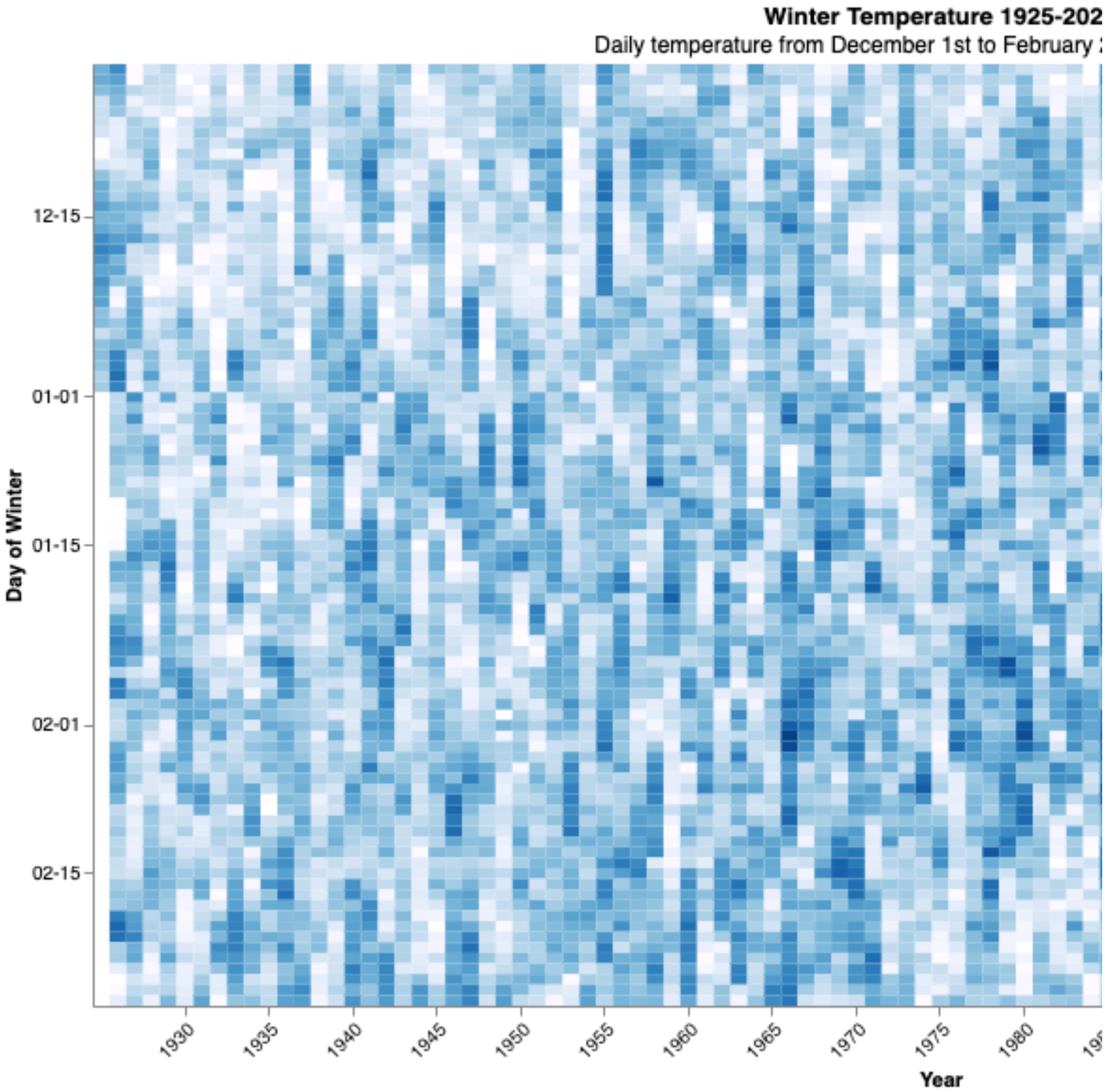
Using a heatmap to look into the daily temperature over the last 99 years can give an insight if the temperature trends have changed. Not onluy can one then look at how the gradient, but also is there are more cyclic trends etc.

In [16]:
```
#Using Vega Altair to plot the heatmap for the summer temperature
summer_heatmap = alt.Chart(summer_temperature,
    title=alt.Title("Summer Temperature 1925-2024",
    subtitle='Daily temperature from June 1st to August 31st over 99 year
    ).mark_rect().encode(
    x=alt.X("year:O", title="Year", axis=alt.Axis(labelAngle=-45,
                                                  values = list(range(192
    y=alt.Y("month_day:N", title="Day of Summer", sort=None, axis=alt.Axi

    color=alt.Color("temperature:Q", scale=alt.Scale(scheme="reds", domai
    tooltip=["year", "month_day", "temperature"]
).properties(
    width=900,
    height=500,
)

summer_heatmap.show()
```



This plot gives much more information than the pervious, where you can see that the summers seem to have more days with higher temperature compared to earlier years. While the overall gradient change is more subtle, the longer periods of low temperature in the summer seem to happen less and less.

In [17]:
```
winter_heatmap = alt.Chart(winter_temperature,
    title=alt.Title("Winter Temperature 1925–2024",
    subtitle='Daily temperature from December 1st to February 28th over 9
    ).mark_rect().encode(
    x=alt.X("year:O", title="Year", axis=alt.Axis(labelAngle=-45,
                                               values = list(range(192
    y=alt.Y("month_day:N", title="Day of Winter", sort=None, axis=alt.Axi

    color=alt.Color("temperature:Q", scale=alt.Scale(scheme="blues", doma
    tooltip=["year", "month_day", "temperature"]
).properties(
    width=900,
    height=500,
)
winter_heatmap.show()
```



Doing the same analysis of the winters in Tromsø is paints a similar picture to the summers. There seems to be less days of very low temperature and if we have a period of low temperature, it doesn't seem to last as long as they did previously. However when comparing late 1920s to the last decade, there seems to be less difference than one might have thought.

Another way of looking at temperature change that is often used when looking at changing climates and temperature, is the temperature anomaly compared to a normal period. We can't use the pre-industrial normal, so a normal that is often used in weather analysis in general is the normal of 1961-1990.

In [ ]:

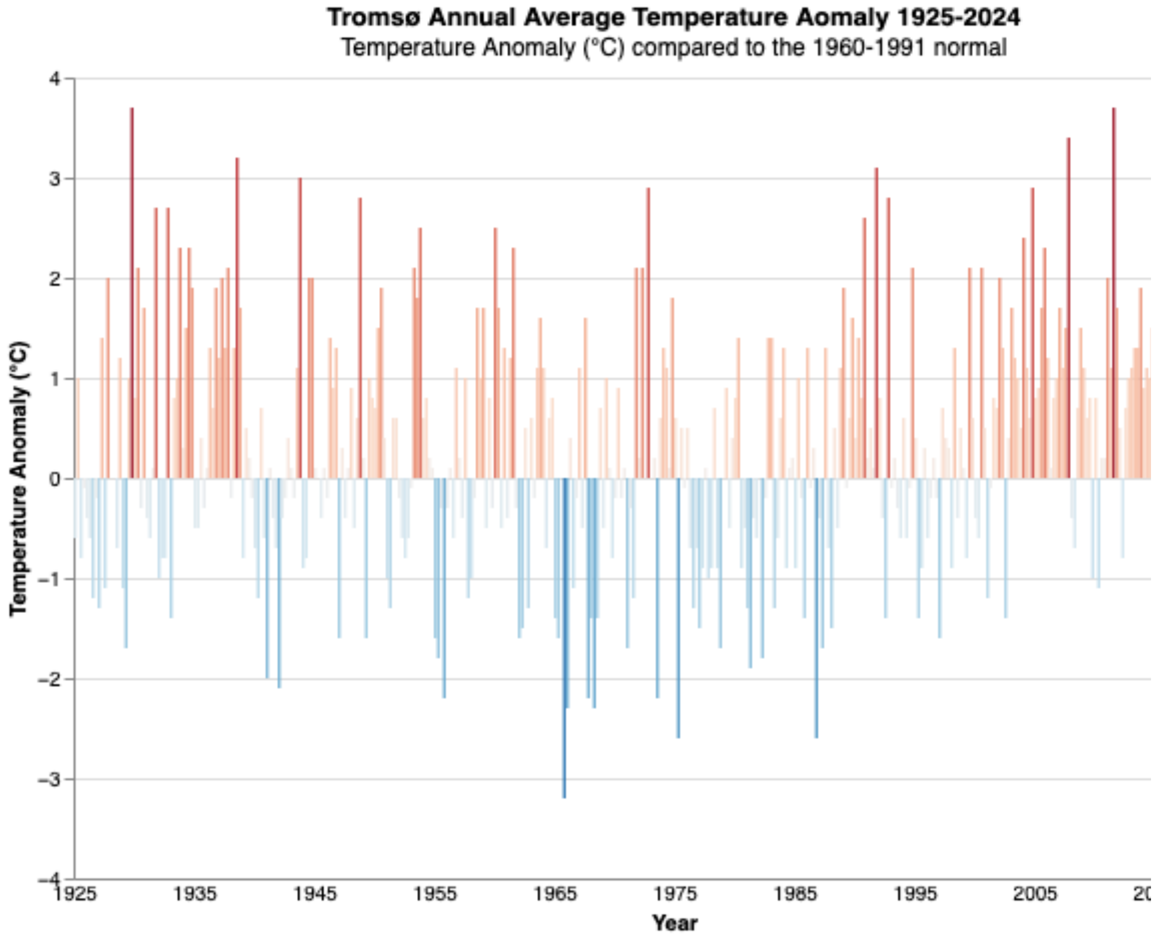
```
#Collecting the temperature anomaly compared to the 1961-1990 normal
amomaly_temperature = vervarslinga.get_temperature_anomaly('1925', '2024'
```

In [19]:

```
#Plotting the normal, with stronger colours used for higher anomalies in
color_change = alt.datum.anomaly > 0

alt.Chart(amomaly_temperature,
title=alt.Title("Tromsø Annual Average Temperature Aomaly 1925-2024",
subtitle='Temperature Anomaly (°C) compared to the 1960-1991 normal')
).mark_bar(
```

```
    bandSize=40,
).encode(
    x = alt.X("year_month:O", axis=alt.Axis(labelAngle=0,
                                            labelExpr='substring(datum.va
                                            values=['1925-03', '1935-03',
                                            labelOverlap=True, ticks=Fals
    y =alt.Y('anomaly:Q', title="Temperature Anomaly (°C)"),
    color=alt.Color('anomaly:Q',  # Use 'value' for coloring
                    scale=alt.Scale(scheme='redblue', domain=[4, -4])),
).properties(
    width=600,
    height=400,
)
```

Out[19]:



This clearly doesn't look like the plot one is used to seeing in newspapers when reading about climate change. Here it is important to note that because Tromsø is so much smaller than the global average, it is not unexpected to see this shirt in trend compared to the global surface temperature anomalies. You can still see that Tromsø follows some of the trends of the global anomalies, with warmer years happening more often and with a higher intensity compared to before.

Summary

It seems as Tromsø is getting warmer, with fewer cold days in the summer and with warmer days in the winter. The overall change compared to the normal also shows that the change is more intense, with some years being as high as up to 3.5 degrees over the normal . While there might have been some individual warm years in history, the trends show that we now can expect this to be the normal if this continues like it has done in the last decade.

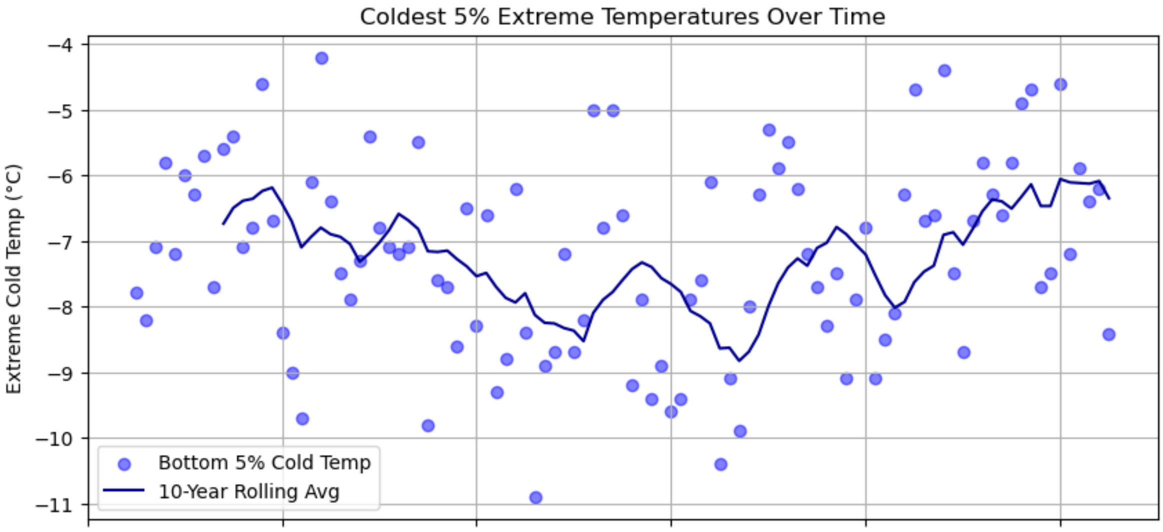
2. The weather has become more extreme, with higher temperatures and stronger winds classified as extreme (top 5%)

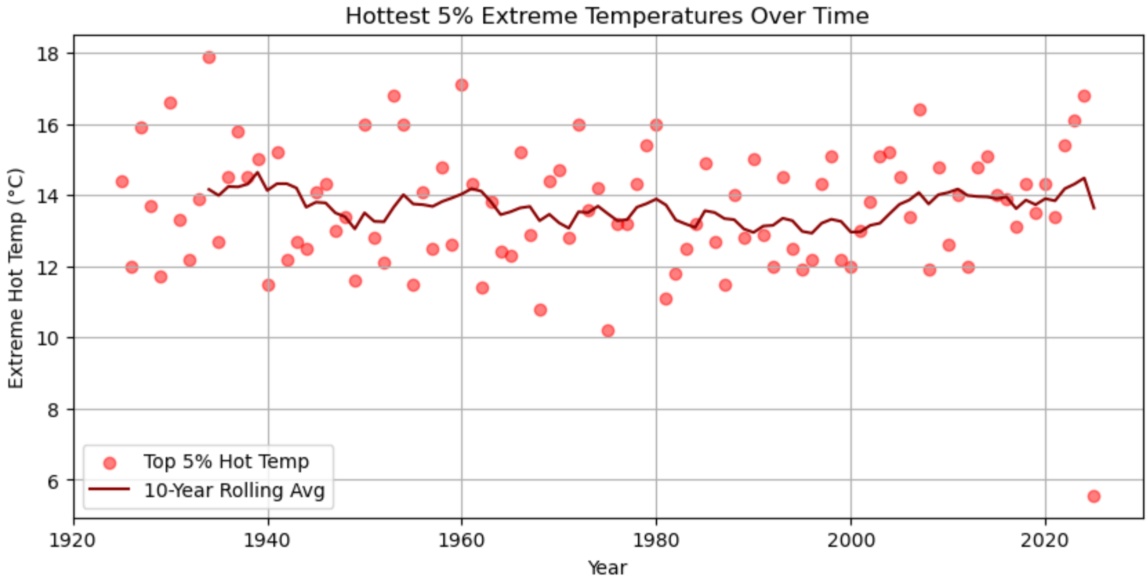
Seeing as the temperature has grown warmer, how are the extreme days? And what about the winds? Do we see more storms now than we did?

In [ ]:
```
#Combining the seaons data into one dataframe for the entire year
daily_temp = (pd.concat([summer_temperature, winter_temperature, atumn_te
                 .assign(date=lambda x: pd.to_datetime(x['date']))
                 .sort_values('date')
                 .reset_index(drop=True))
```

In [21]:
```
#Collecting the upper 5% and lower 5% of the temperature for each year to
extreme_temps = daily_temp.groupby("year").agg(
    extreme_hot=("temperature", lambda x: x.quantile(0.95)),
    extreme_cold=("temperature", lambda x: x.quantile(0.05))
).reset_index()
```

In [22]:
```
# Create a figure with two subplots showing each extreme
fig, axes = plt.subplots(2, 1, figsize=(10, 10), sharex=True)

axes[0].scatter(extreme_temps["year"], extreme_temps["extreme_cold"], col
axes[0].plot(extreme_temps["year"], extreme_temps["extreme_cold"].rolling
axes[0].set_ylabel("Extreme Cold Temp (°C)")
axes[0].set_title("Coldest 5% Extreme Temperatures Over Time")
axes[0].legend()
axes[0].grid(True)

axes[1].scatter(extreme_temps["year"], extreme_temps["extreme_hot"], colo
axes[1].plot(extreme_temps["year"], extreme_temps["extreme_hot"].rolling(
axes[1].set_ylabel("Extreme Hot Temp (°C)")
axes[1].set_xlabel("Year")
axes[1].set_title("Hottest 5% Extreme Temperatures Over Time")
axes[1].legend()
axes[1].grid(True)

plt.show()
```





Compared to the seasonal change where we conluded that the winters and summers are getting warmer, we do still see that we have cold days in Tromsø, though less compared to the 1980s when it was at its coldest. The extreme days for the warm side seems to be more stable, meaning that our summers are just warmer in general, but doesn't seem to have a big increase in extreme warmth (yet). There are hints that this might be happening in this and the upcoming decades however.

In Norway we classify storms based on wind strengths, so we will now see if we have more storms and stronger winds in recent time compared to historical data.

In [23]:
```
#Collecting the max wind data from the Vervarslinga station from 1960-202
daily_wind = vervarslinga.get_daily_max_wind('1960-01-01', '2025-02-28')
daily_wind
```

Out[23]:

	date	max_wind_speed
0	1960-01-01	8.2
1	1960-01-02	7.7
2	1960-01-03	4.1
3	1960-01-04	0.0
4	1960-01-05	0.0
...	...	...
23783	2025-02-23	7.6
23784	2025-02-24	7.3
23785	2025-02-25	7.3
23786	2025-02-26	6.4
23787	2025-02-27	5.2

23788 rows × 2 columns

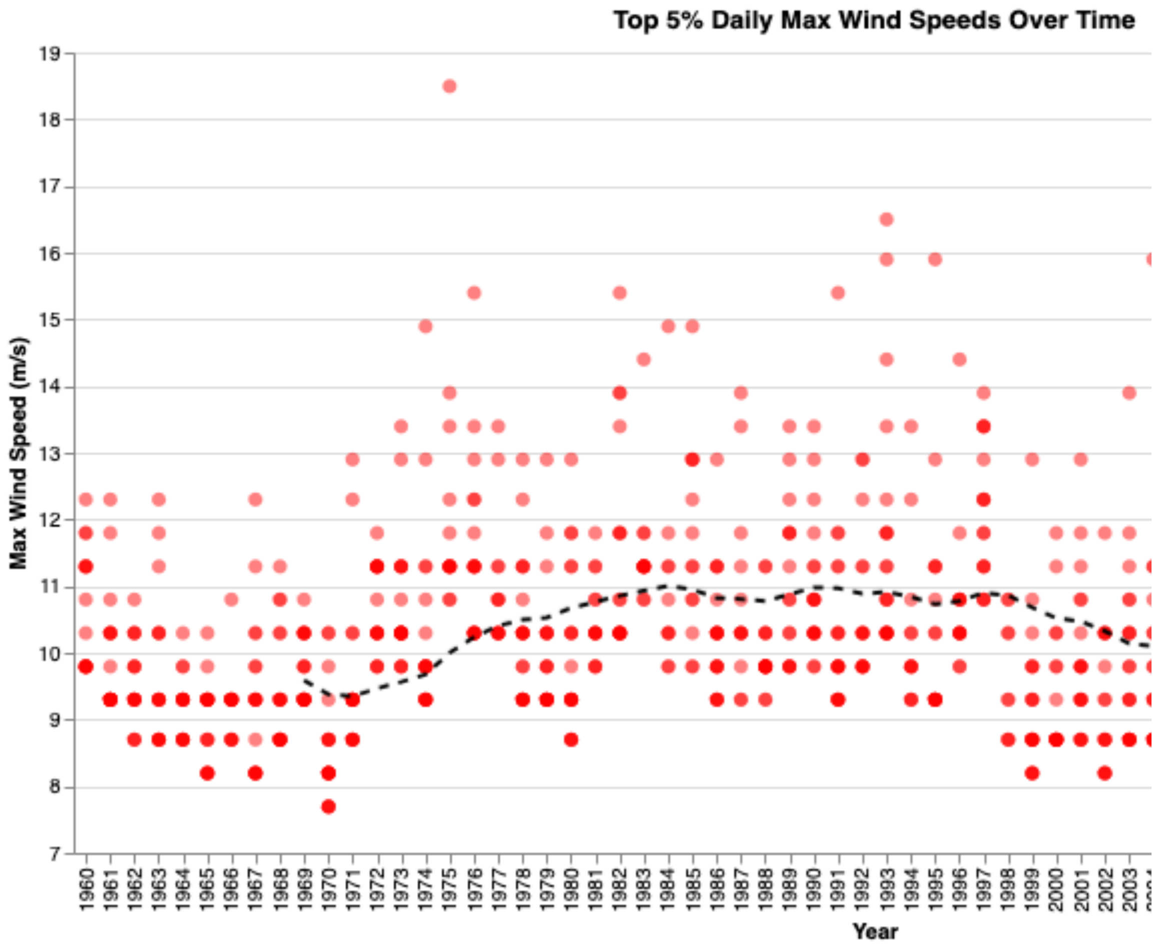
In [24]:

```
#Reformatting the dataframes to make them easier to work with and finding
daily_wind['year'] = pd.to_datetime(daily_wind['date']).dt.year
precentile = daily_wind.groupby('year').quantile(0.95).reset_index().drop
precentile.rename(columns={'max_wind_speed': '95th_percentile_wind_speed'

df_extreme = daily_wind.merge(precentile, on='year')
df_extreme = df_extreme[df_extreme['max_wind_speed'] > df_extreme['95th_p
```

In [25]:

```
#Create the scatter plot for top 5% wind speeds
scatter = alt.Chart(df_extreme).mark_circle(size=50, opacity=0.5, color="
    x=alt.X("year:O", title="Year"),  # Discrete years
    y=alt.Y("max_wind_speed:Q", title="Max Wind Speed (m/s)", scale=alt.S
    tooltip=["date", "max_wind_speed"]
)

#Compute 10-year rolling average for trend line
df_trend = df_extreme.groupby("year")["max_wind_speed"].mean().rolling(10

#Create the trend line
trend = alt.Chart(df_trend).mark_line(color="black", strokeDash=[5,5]).en
    x=alt.X("year:O"),
    y=alt.Y("max_wind_speed:Q", title="Max Wind Speed (m/s)")
)

#Combine both plots
alt.layer(scatter, trend).properties(
    title="Top 5% Daily Max Wind Speeds Over Time",
    width=800,
    height=400
)
```

Out[25]:



Overall it does seem that we do have stronger winds now compared to when we started collecing the data, but when looking at the data, there is a clear change between before and after 2004. This might come from more senstivie or better equipment to measure data, and will have an impact on the analysis. Another anomaly in the data is the day in February 1975. Looking at local newspapers, we can conlude a large storm was in Tromsø at that time, creating damages in the millions of kroner at the time and even blowing a boat up onto a beach. The newspaper in question can be found in the "docs" folder of this project.

Summary

There seems to be some changes in the extremes of the weather in Tromsø, most visible in the wind strenghts, however you can also see trend changes in the temperatures as well.

3. There exists some local differences between the different sensors, especially depending on the geography

In this we will only look at the last 20 years of data. To make sure this project isn't too big, we will only look at wind as it is clearly dependent on geography. As a reminder, the stations we can use are;

In [26]:
```
chosen_stations
```

Out[26]:

	station_id	name	date	year	lon	lat	geometry
0	SN90450	Tromsø (Ververslinga)	1895-08-01	1895	18.9368	69.6537	POINT (18.9368 69.6537)
1	SN90490	Tromsø LH	1964-09-30	1964	18.9133	69.6767	POINT (18.9133 69.6767)
2	SN90400	Tromsø (Holt)	1987-05-04	1987	18.9095	69.6538	POINT (18.9095 69.6538)
4	SN90720	Måsvik	2003-10-08	2003	18.6938	69.9903	POINT (18.6938 69.9903)
3	SN90491	Stor-Kjølen	1998-04-02	1998	18.7875	69.7333	POINT (18.7875 69.7333)

In [27]:
```
#Makes a list of the chosen stations for easier to ge the data
chosen_station_id = chosen_stations['station_id'].tolist()
chosen_station_name = chosen_stations['name'].tolist()
```

In [28]:
```
#Creating a dict to collect the information easier
compare_station = {}
for name, st_id in zip(chosen_station_name, chosen_station_id):
    compare_station[name] = rd.ReadingData(st_id, client_id, client_secre
    print(f"Getting data for station {name}")
    compare_station[name] = compare_station[name].get_daily_max_wind('201
```

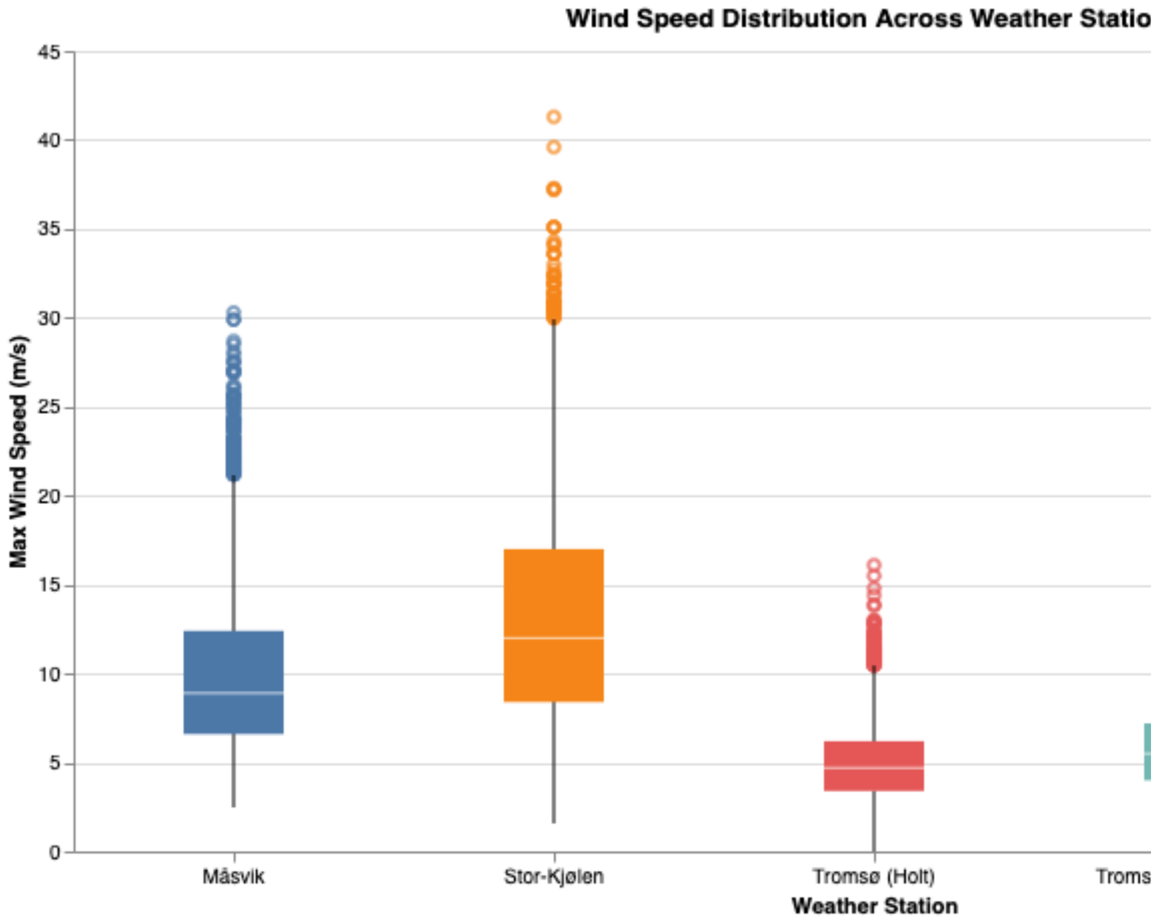
```
Getting data for station Tromsø (Vervarslinga)
Getting data for station Tromsø LH
Getting data for station Tromsø (Holt)
Getting data for station Måsvik
Getting data for station Stor-Kjølen
```

In [29]:
```
#Reformatting the dataframes to make them easier to work with and to be c
df_list = []

for station, df in compare_station.items():
    df = df.copy()
    df["station"] = station
    df_list.append(df)


df_all = pd.concat(df_list, ignore_index=True)
```

In [30]:
```
boxplot = alt.Chart(df_all).mark_boxplot(size=50).encode(
    x=alt.X("station:N", title="Weather Station",
            axis=alt.Axis(labelAngle=0)),  # Rotate labels
    y=alt.Y("max_wind_speed:Q", title="Max Wind Speed (m/s)"),
    color=alt.Color("station:N", legend=None)  # Color-coded per station
).properties(
    title="Wind Speed Distribution Across Weather Stations",
    width=800,
    height=400
```

```
)

boxplot
```

Out[30]:



Looking at the max wind distributions noe can see there are clear local differences. Stor-Kjølen has the most wind, and (not surprisingly) is at the most open station, being on top of a mountain above where trees grow. Måsvik is open to the Atlantic Ocean and wil thus also have more wind, while Holt and Vervarslinga are more protected by both buildings, but also montains surrounding Tromsø.

Conclusion

Looking at the weather and observations in Tromsø does paint a picture of a trend that things are getting warmer and with more extreme winds. Like the rest of the world, climate change seems to have an impact and will probably continue to affect our daily weather. Given more time, a look into precipitation, snow depth and cloud cover could also have been interesting variables to look at and to see if they show the same story as we have seen now.